INCH-POUND

MIL-T-27/9B
9 November 2009
SUPERSEDING
MIL-T-27/9A
6 December 1976

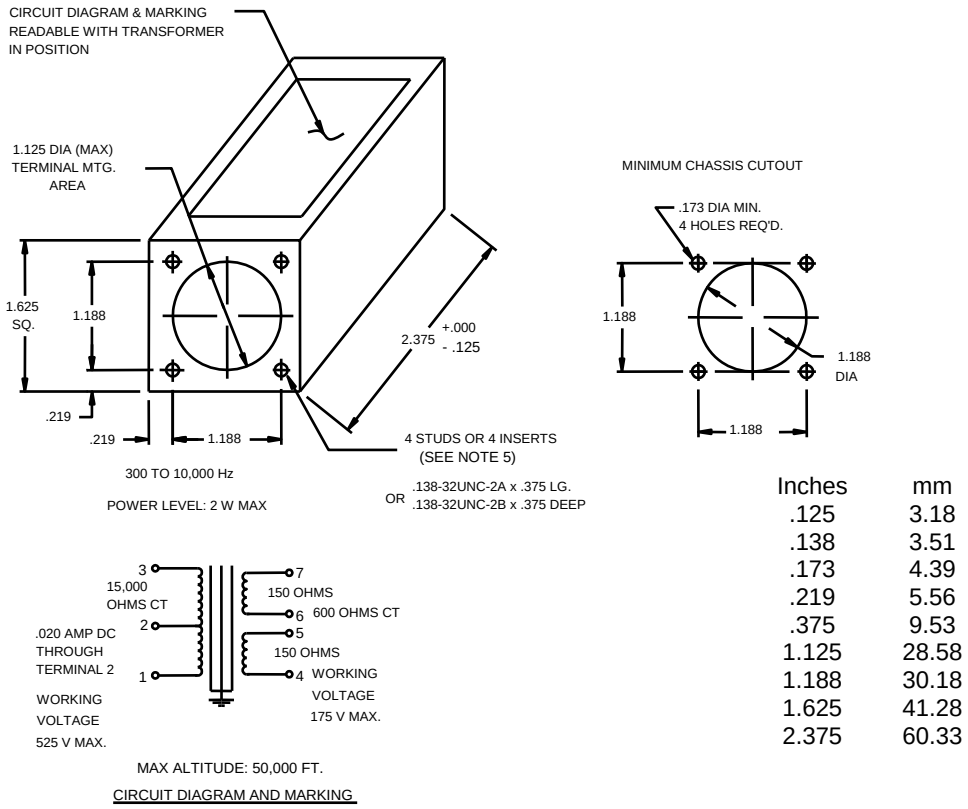
MILITARY SPECIFICATION SHEET

# TRANSFORMER, AUDIO FREQUENCY

Inactive for new design
after 25 August 1998

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.




CIRCUIT DIAGRAM AND MARKING

| Inches | mm |
|---|---|
| .125 | 3.18 |
| .138 | 3.51 |
| .173 | 4.39 |
| .219 | 5.56 |
| .375 | 9.53 |
| 1.125 | 28.58 |
| 1.188 | 30.18 |
| 1.625 | 41.28 |
| 2.375 | 60.33 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance on overall case dimensions is +.000 (0.00 mm) .063 (1.60 mm)
4. Tolerance on mounting dimensions is +.016 (0.41 mm).
5. Inserts not recommended for new design.
6. Terminal arrangement optional within mounting area indicated.

FIGURE 1. Dimensions and configurations.

AMSC N/A FSC 5950

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REQUIREMENTS: (When numbers in parentheses, such as (1-3) are used, they indicate the winding and the extreme terminals of the winding. When the extreme terminals of the two windings are used, (4-7), the windings are connected in series, terminals 5 and 6 are connected.)

Electrical ratings:

Primary power level: 2 W, maximum; .1 W, minimum.

Working voltage:

(1-3): 525 V, maximum.
(4-5): 175 V, maximum.
(6-7): 175 V, maximum.

Primary current: .020 ampere dc through terminal 2, .001 ampere dc, maximum unbalance.

Primary impedance (1-3): 15,000 ohms ct.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal encased.

Material: Steel.

Terminals: Solder terminals, No. 18 AWG.

Height: .375 (9.53 mm) +.000 -.188 (4.78 mm).

Weight: .72 pound, maximum.

Operating temperature range: -55°C to +105°C.

Altitude: 50,000 feet, maximum.

Life expectancy: 10,000 hours, minimum.

Terminal strength: MIL-STD-202, method 211, test condition A, 5.0 pounds.

Dielectric withstanding voltage: MIL-STD-202, method 301

At sea level: 1,500 V rms.
At reduced barometric pressure: 500 V rms.

Electrical characteristics:

Harmonic distortion: Not more than 1 percent with 173 V, 300 Hz across (1-3) and .020 ampere dc through terminal 2.



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No load (center-tap voltage unbalance only): With 173 V, 1,000 Hz across (1-3) and .020 ampere dc through terminal 2:

Unbalance (4-5) and (6-7): 1 percent, maximum.
Unbalance (1-2) and (2-3): 1 percent, maximum.

Secondary load impedance: (4-5) – 150 ohms.
(6-7) – 150 ohms ±10 percent.
(4-7) – 600 ohms ct.

Primary impedance: With approximately 30 V, 1,000 Hz across (1-3) and .020 ampere dc though terminal 2 and 600 ohms ct across (4-7), the primary impedance shall be 15,000 ohms ct ±10 percent across (1-3).

Frequency response:

Range: 300 to 10,000 Hz at rated power, flat within ±2 dB, reference frequency = 1,000 Hz; test voltage 30 V rms (1-3).

Insertion loss: 1.5 dB maximum, with 173 V, input at 1,000 Hz across (1-3).

Resonance: Second resonant frequency = 20,000 Hz, minimum.

Polarity: Additive with terminals 3 and 4, and 5 and 6 connected.

Temperature rise: 20°C maximum, above an ambient of 85°C, with rated power at 300 Hz.

Vibration (high frequency): MIL-STD-202, method 204, test condition B.

Shock (specified pulse): MIL-STD-202, method 213, test condition I.

Part or Identification Number (PIN): M27/9-01A or -01B.

TABLE I. Cross reference.

| Mounting | MS90006B | M27/9 |
|---|---|---|
| | Dash no. | Dash no. |
| Studs | -A | -01A |
| Inserts | -B | -01B |

NOTE: Qualification approval to MIL-T-27/9 shall be sufficient to grant qualification approval to MIL-T-27/3 through MIL-T-27/8.



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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-008)

Review activities:
Army - AR, CR4
Navy - AS, MC
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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